THE COUNT OF GEOMETRICALLY IRREDUCIBLE FIBRE COMPONENTS IS CONSTANT OVER AN OPEN OF THE CLOSURE OF A POINT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7740-7749 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 7751-7791, the complete author proof (39 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Constructibility of the count of geometrically irreducible fibre components and the generic-fibre behaviour of the count; the spreading argument is retained.

Source reference. Original lemma lemma-base-change-fibres-nr-geometrically-irreducible-components: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7453–7466. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-cover-generic-fibre-neighbourhood: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6658–6666. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-geom-irreducible-generic-fibre: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7557–7564. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-irreducible-components-in-neighbourhood: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7393–7399. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-make-components-generic-fibre-geometrically-irreducible: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6772–6792. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma morphisms-lemma-etale-open: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7791–7791. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma properties-lemma-characterize-integral: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7791–7791. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $n_{X/Y}$ be the function on Y counting the numbers of geometrically irreducible components of fibres of f introduced in Lemma lemma-base-change-fibres-nr-geometrically-irreducible-components. Assume f of finite type. Let $y \in Y$ be a point. Then there exists a nonempty open $V \subset \overline{\{y\}}$ such that $n_{X/Y}|_V$ is constant.*

Proof. Let Z be the reduced induced scheme structure on $\overline{\{y\}}$. Let $f_Z : X_Z \to Z$ be the base change of f. Clearly it suffices to prove the lemma for f_Z and the generic point of Z. Hence we may assume that Y is an integral scheme, see Properties, Lemma properties-lemma-characterize-integral. Our goal in this case is to produce a nonempty open $V \subset Y$ such that $n_{X/Y}|_V$ is constant.

We apply Lemma lemma-make-components-generic-fibre-geometrically-irreducible to $f : X \to Y$ and we get $g : Y' \to V \subset Y$. As $g : Y' \to V$ is surjective finite étale, in particular open (see Morphisms, Lemma morphisms-lemma-etale-open), it suffices to prove that there exists an open $V' \subset Y'$ such that $n_{X'/Y'}|_{V'}$ is constant, see Lemma lemma-base-change-fibres-nr-geometrically-irreducible-components. Thus we see that we may assume that all irreducible components of the generic fibre X_η are geometrically irreducible over $\kappa(\eta)$.

At this point suppose that $X_\eta = X_{1,\eta} \bigcup \ldots \bigcup X_{n,\eta}$ is the decomposition of the generic fibre into (geometrically) irreducible components. In particular $n_{X/Y}(\eta) = n$. Let X_i be the closure of $X_{i,\eta}$ in X. After shrinking Y we may assume that $X = \bigcup X_i$, see Lemma lemma-cover-generic-fibre-neighbourhood. After shrinking Y some more we see that each fibre of f has at least n irreducible components, see Lemma lemma-irreducible-components-in-neighbourhood. Hence $n_{X/Y}(y) \geq n$ for all $y \in Y$. After shrinking Y some more we obtain that $X_{i,y}$ is geometrically irreducible for each i and all $y \in Y$, see Lemma lemma-geom-irreducible-generic-fibre. Since $X_y = \bigcup X_{i,y}$ this shows that $n_{X/Y}(y) \leq n$ and finishes the proof. $\square$